PROJECT REPORT

ServiceNow-Based Employee Laptop Request Application

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1. INTRODUCTION

1.1 Project Overview

The ServiceNow-Based Employee Laptop Request Application is a centralized self-service workflow for employees who need a laptop for their work. It replaces informal requests made through email or verbal communication with a structured request record, manager approval, status tracking, and email notifications. The application is built on the ServiceNow platform using a request form/Record Producer, a custom Laptop Request Form table, Flow Designer, Business Rules, and platform approval and notification capabilities.

The main custom table is Laptop Request Form (u_laptop_request_form). The solution also uses the built-in User table and the sysapproval_approver table for approval records. A request number is generated automatically in the REQ-plus-digits format, so the employee does not need to enter it manually.

1.2 Purpose

- Provide one consistent place for employees to submit laptop requests.
- Capture the business reason, requested date/time, request type, and selected manager.
- Classify requests as Basic or Advanced to make the request type explicit.
- Prevent the same requester from creating more than one request in the same calendar year, while allowing a request when the previous one belongs to a different calendar year.
- Route requests to the relevant manager and record the approval decision.
- Keep employees and relevant stakeholders informed through status changes and email notifications.
- Improve accountability and visibility compared with email or verbal follow-ups.

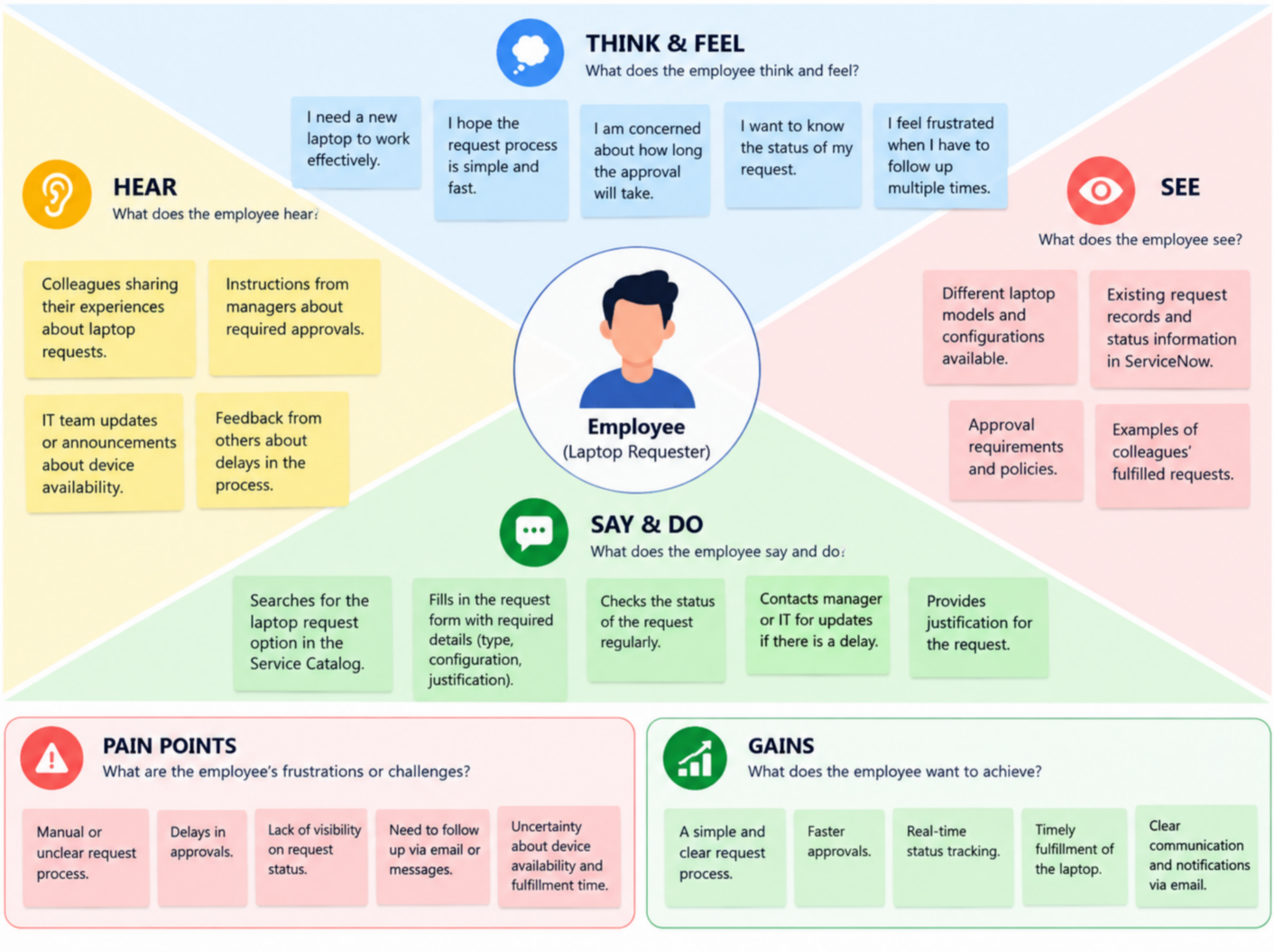
2. IDEATION PHASE

2.1 Problem Statement

Employees may request laptops through scattered channels such as emails or verbal conversations. These channels can make it difficult to standardize request details, identify the current status, confirm manager approval, and detect repeated requests. The organization needs a centralized and controlled process that captures consistent information, applies duplicate-request rules, routes approval, and communicates outcomes.

How might we provide employees with a simple, traceable laptop-request process while ensuring that each request contains the required information, is checked against existing requests for the same calendar year, and follows a manager-approval workflow?

2.2 Empathy Map Canvas



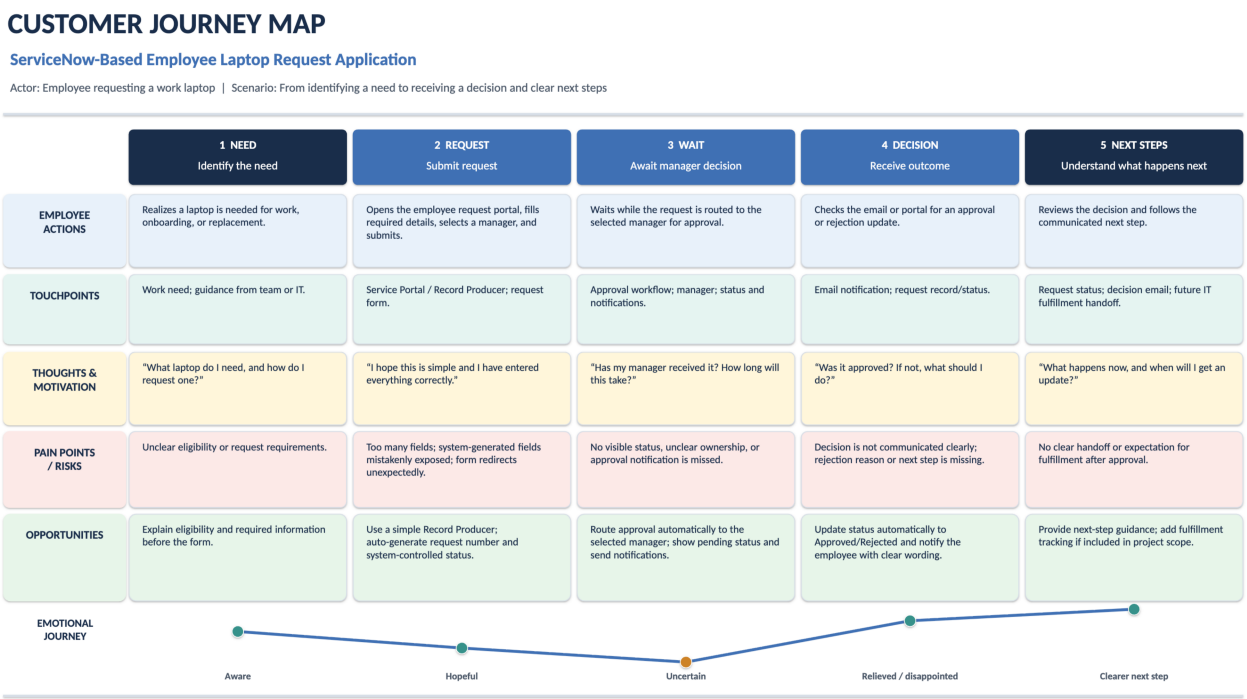
2.3 Brainstorming

Idea	Purpose / decision
Centralized request form	Collect the same core information for every request.
Automatic request number	Give each request a unique reference without manual entry.
Basic / Advanced classification	Make the category of laptop request explicit.
Required-field validation	Stop incomplete submissions and identify missing fields.
Year-based duplicate check	Block a same-requester duplicate within the same

	calendar year, while permitting a request from a different year.
Manager approval through Flow Designer	Create and track an approval record for the selected manager.
Email notifications	Communicate request or approval outcomes to the relevant person.
Role, group, and ACL configuration	Restrict access according to the intended user roles and permissions.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

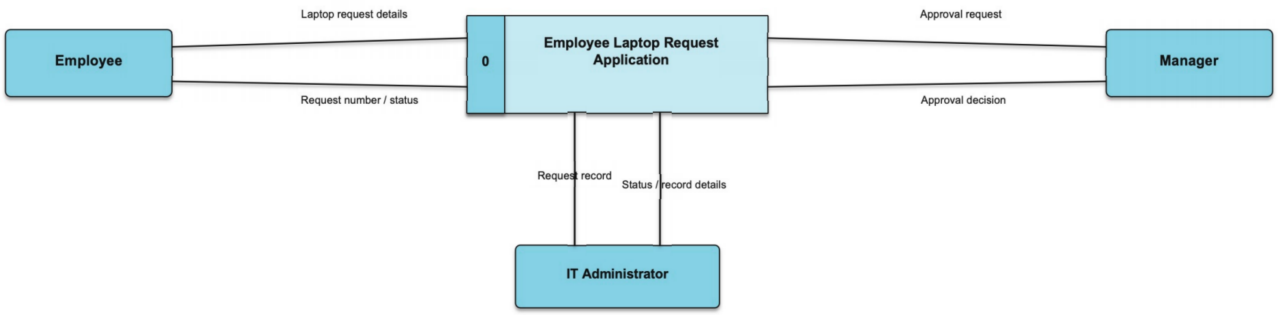


3.2 Solution Requirement

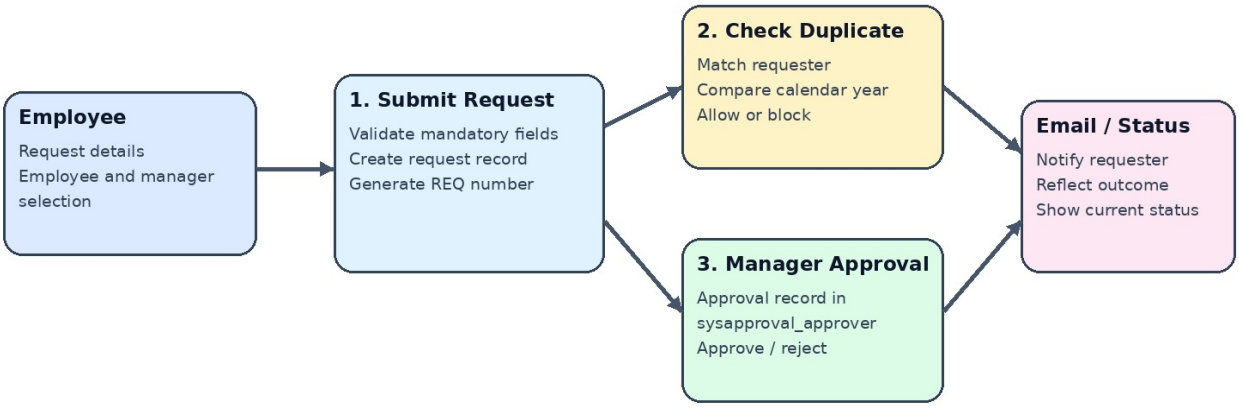
ID	Requirement	Type	Evidence / status
FR-01	Provide a centralized laptop request form.	Functional	Implemented in ServiceNow.
FR-02	Capture employee name, email, needed date/time, short description, type, reason, and manager.	Functional	Fields confirmed in project setup.
FR-03	Validate mandatory fields.	Functional	UAT-02 passed: incomplete submission was blocked and missing field indicated.
FR-04	Generate a request number automatically.	Functional	UAT-01 passed; number uses REQ plus digits.
FR-05	Check same-requester requests within the calendar year.	Functional	Business Rule described and tested.
FR-06	Allow a new request if the existing request is from a different calendar year.	Functional	UAT-17 passed.
FR-07	Create a manager approval record.	Functional	Flow Designer and sysapproval_approver are used.
FR-08	Reflect approval/rejection in request status.	Functional	Rejection-to-Rejected outcome was recorded as passed in prior UAT notes.
FR-09	Send requester email notifications for relevant outcomes.	Functional	Requester notification and rejection email outcomes were recorded as passed.
FR-10	Control access through roles/groups/ACLs.	Security	Configuration confirmed; unauthorized-access test recorded as passed.
NFR-01	Keep requests traceable and consistently structured.	Quality	Supported by request records and generated numbers.

3.3 Data Flow Diagram

Data flow diagram (Level 0)



Data flow diagaram(Level 1)



Level 1 view: employee details flow into request validation and record creation. The request is checked for duplicates, then routed through manager approval. The system updates the request outcome and sends configured notifications. This is a logical representation of the workflow, not a captured ServiceNow execution trace.

3.4 Technology Stack

Technology / component	Role in the project
ServiceNow platform	Application platform for forms, records, access controls, and workflow.
Record Producer / request form	Employee-facing entry point for submitting a laptop request.
Custom table: u_laptop_request_form	Stores laptop request details and status.
Flow Designer	Orchestrates the manager-approval process.
sysapproval_approver	Built-in table used to store approval records.
Business Rules / server-side scripting	Enforces the calendar-year duplicate-request rule.
Notifications / email	Communicates relevant request and approval outcomes.
User table	Provides user/manager references.
Roles, groups, ACLs	Support controlled access to records and actions.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution element	Expected benefit
Requests are scattered or informal.	Central ServiceNow request form and record.	Consistent intake and traceability.
Request details may be incomplete.	Mandatory-field validation.	Fewer incomplete submissions.
Repeated requests can be hard to identify.	Business Rule checks requester and calendar year.	Reduces same-year duplicate submissions.
Approval ownership is unclear.	Flow Designer with manager approval record.	Approval is linked to the request.
Employees need outcome updates.	Status updates and email notifications.	Less manual follow-up.
Access should be controlled.	Roles, groups, and ACLs.	Restricts unauthorized access.

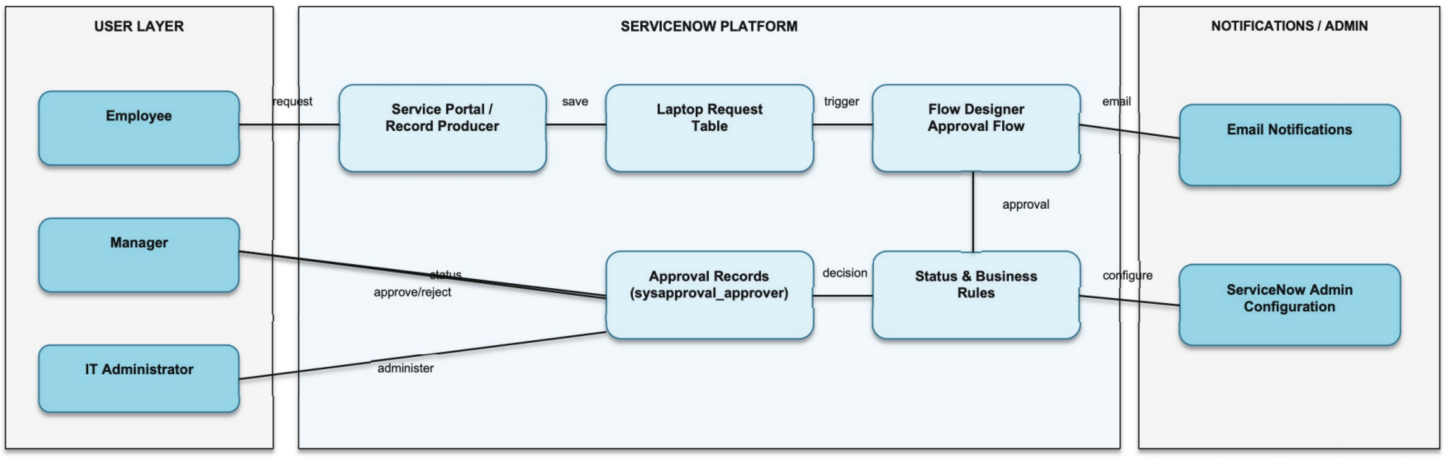
4.2 Proposed Solution

The proposed solution is a ServiceNow-based workflow that begins when an employee submits the laptop request form. The form collects the required information and creates a record in u_laptop_request_form. The request number is generated automatically. A Business Rule checks whether the requester already has a request in the same calendar year; a same-year duplicate is blocked, while a request whose earlier record belongs to another calendar year is accepted. Flow Designer creates the manager-approval record in sysapproval_approver and supports the approval outcome. The request status and configured notifications communicate the result.

4.3 Solution Architecture

The architecture diagram summarizes the main logical components. Actual flow steps and notification timing should be verified against the final ServiceNow configuration before submission.

Architecture Diagram: Employee Laptop Request and Manager Approval



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Deliverable / checkpoint
1. Ideation	Identify employee pain points; define the problem; prepare empathy map and brainstorm ideas.	Problem statement and ideation artifacts.
2. Requirement analysis	Identify request fields, validation rules, approval needs, notifications, and access requirements.	Requirement list and journey map.
3. Data model and form	Create the custom request table and request form/Record Producer; configure fields and request number.	Working request intake.
4. Business rules	Implement same-requester, same-calendar-year duplicate validation.	Duplicate validation behavior.
5. Approval workflow	Configure Flow Designer and approval records for the selected manager.	Manager approval workflow.
6. Notifications and access	Configure email notifications, roles, groups, and ACLs.	Notifications and access controls.
7. Testing	Run functional/UAT cases for form validation, numbering, duplicate logic, access, approvals, and email.	UAT outcomes and issue list.
8. Documentation	Capture actual screenshots, confirm links, and prepare final report.	Submission-ready report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

User Acceptance Testing was carried out in the ServiceNow development instance. The supplied UAT record contains 18 test cases covering request submission, automatic request numbering, required-field validation, saved request details, manager approval and rejection, annual duplicate prevention, role-based access, catalog availability, and email notifications. All 18 test cases are recorded as Pass in the supplied UAT, with TC-003 marked Pass* because the email field was not separately verified.

The recorded results show that the tested application functions worked as expected in the defined scenarios. Requests were created with generated numbers, incomplete submissions were blocked, manager approvals and rejections updated request status, same-year duplicate requests were blocked, different-year requests were accepted, configured access restrictions were enforced, and manager/requester notifications were received. The UAT records that no defects were reported during the recorded testing.

UAT Results Summary

TEST CASE	SCENARIO	ACTUAL RESULT (AS RECORDED IN UAT)	RESULT
TC-001	Submit a valid laptop request	Request created successfully; request number generated automatically.	Pass
TC-002	Required-field validation	Submission was blocked and the missing required field was indicated.	Pass
TC-003	Request details saved correctly	Request type/reason matched selections; requester/manager were correct; short description and date were correct; details persisted. Email field was not separately verified.	Pass
TC-004	Manager approval record	Approval record created for the selected manager.	Pass
TC-005	Manager approves request	Request status changed to Approved after approval.	Pass
TC-006	Manager rejects request	Manager rejected the request and status changed to Rejected.	Pass
TC-007	Duplicate request in same year	Duplicate submission was blocked and a message was displayed.	Pass
TC-008	Request in a different calendar year	New request was accepted for a requester whose existing request was from a different calendar year.	Pass
TC-009	Employee restricted access	Employee was prevented from accessing the restricted function or record.	Pass
TC-010	Manager approval access	Manager could access the approval record and take action.	Pass
TC-011	Manager notification	Manager received an email with correct request details. No separate email-link functionality	Pass

TEST CASE	SCENARIO	ACTUAL RESULT (AS RECORDED IN UAT)	RESULT
		test is included.	
TC-012	Requester notification	Requester received the expected email notification.	Pass
TC-013	Rejection notification	Requester received an email reflecting the rejection.	Pass
TC-014	Request record/status consistency	Request details and final status were saved correctly.	Pass
TC-015	Catalog item availability	Laptop Request item was visible and the form opened successfully.	Pass
TC-016	Unique request numbers	Both requests received different, automatically generated numbers.	Pass
TC-017	Employee views own completed request	Employee could view their completed request.	Pass
TC-018	Other employee's restricted request	Employee could not access another employee's restricted request.	Pass

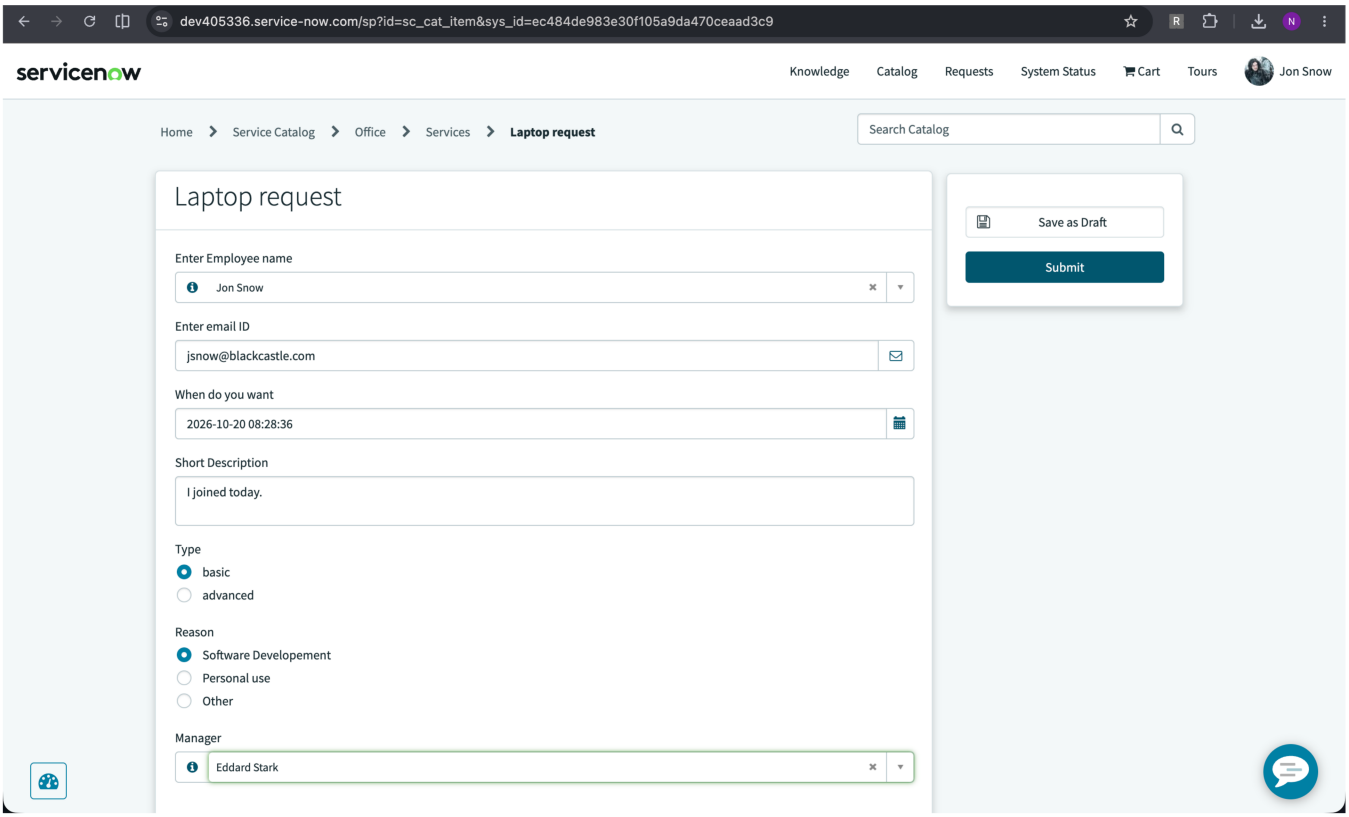
7. RESULTS

The confirmed functional results show that the request intake process can create a record with an automatically generated request number, reject incomplete submissions, apply the calendar-year duplicate rule, route approval, restrict unauthorized access, and send requester email updates. In the recorded tests, a request from a different calendar year was accepted and the initial status before approval was Pending Approval.

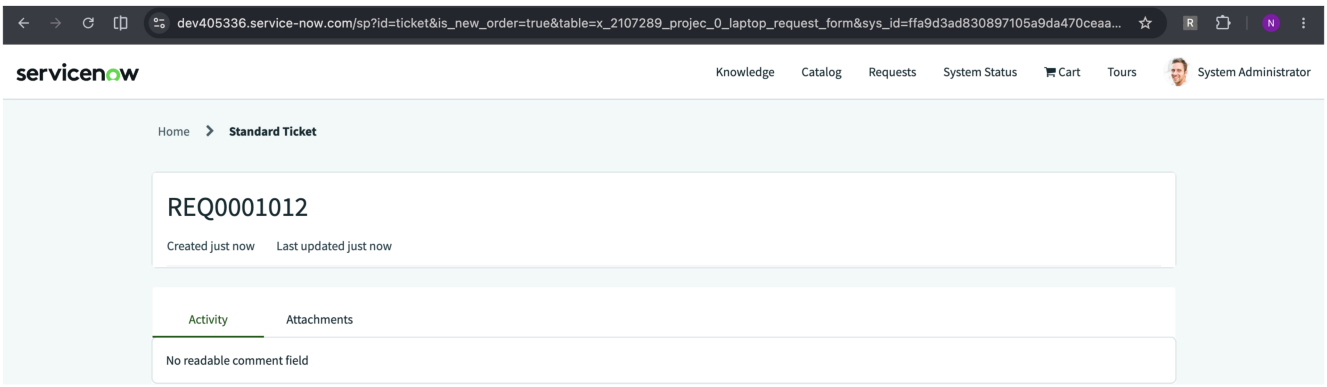
7.1 Output Screenshots

Insert genuine screenshots from the configured ServiceNow instance in the spaces below. Avoid using mock screenshots as evidence.

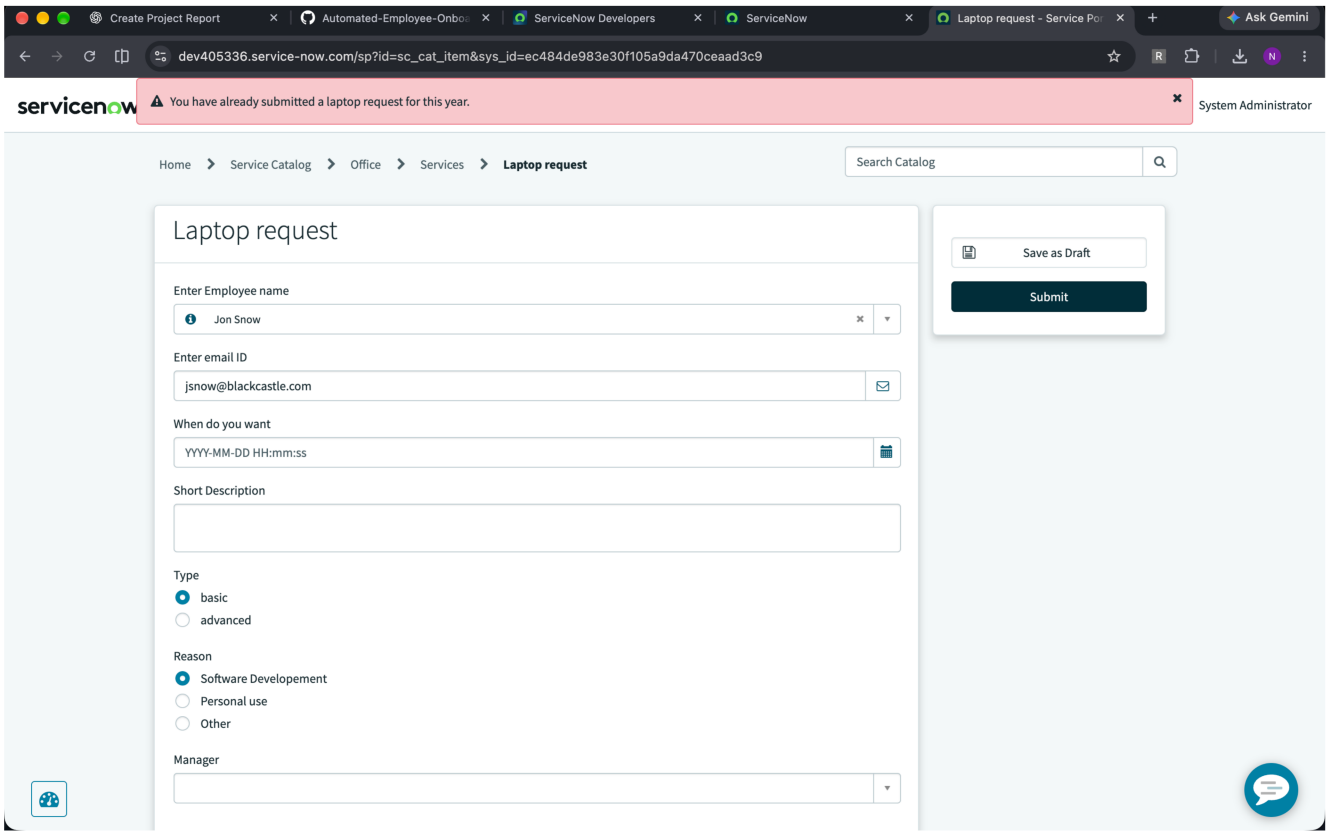
Screenshot 1: Employee laptop request form with required fields



Screenshot 2: Successful submission showing the generated REQ number

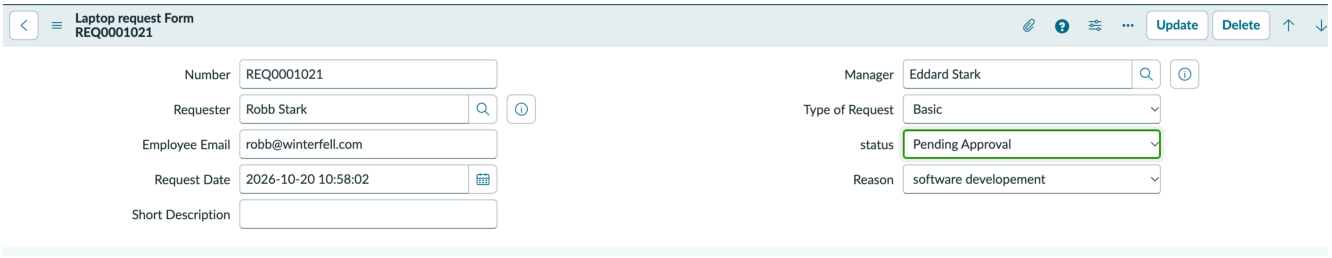


Screenshot 3: Same-year duplicate request blocked



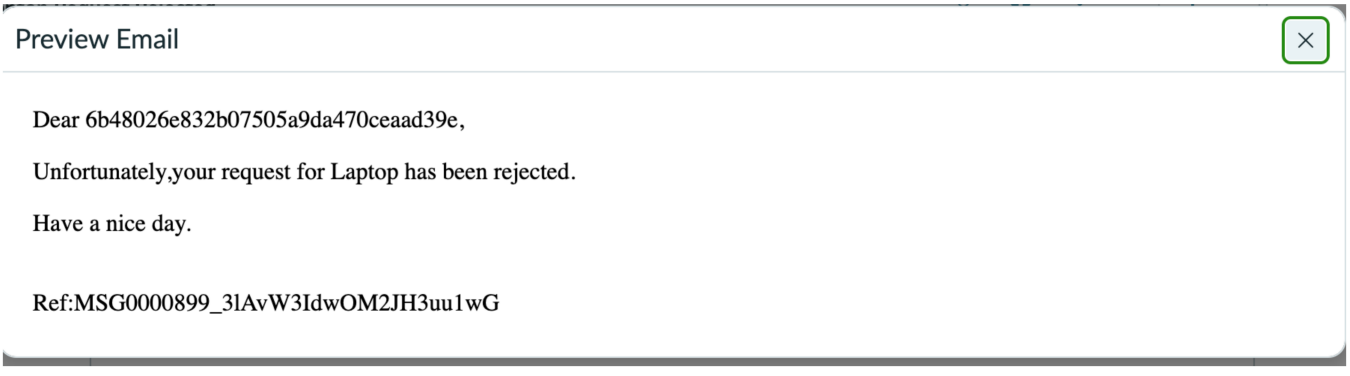
Screenshot 4: Approval record / manager approval view

Screenshot 5: Request status before approval (Pending Approval)



Screenshot 6: Requester email after Approval

Screenshot 7: Requester email after Rejection



8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralizes laptop requests in a single platform.
- Improves traceability through a generated request number and stored request record.
- Uses required-field validation to reduce incomplete submissions.
- Applies a defined duplicate-request rule based on the calendar year.
- Provides a structured manager-approval path.
- Supports email communication and clearer request outcomes.
- Uses ServiceNow access-control capabilities to limit unauthorized access.

Disadvantages / limitations

- The workflow depends on correct configuration of fields, references, approval routing, and notification recipients.
- Approval turnaround still depends on the manager taking action.
- Email delivery and timing depend on ServiceNow notification configuration and the email environment.
- The current evidence does not include quantitative performance or scalability measurements.
- Formal usability feedback from a broader group of employees was not documented in the available notes.

9. CONCLUSION

The ServiceNow-Based Employee Laptop Request Application provides a structured alternative to informal laptop requests. It combines a centralized form, automatic request numbering, mandatory-field validation, a calendar-year duplicate check, manager approval, access controls, status handling, and email notifications. The functional outcomes recorded in the project conversations support the core workflow, including rejection of incomplete and same-year duplicate submissions, acceptance of a request from a different calendar year, and notification of the requester after rejection. The project meets its main goal of making laptop requests more controlled and traceable.

10. FUTURE SCOPE

- Add a dashboard for employees and IT teams to view requests by status, type, and age.
- Introduce reminders and escalation rules for approvals that remain pending beyond a configured period.
- Integrate request fulfilment with inventory/asset management so available laptops can be checked before approval or allocation.
- Add reporting for request volume, approval time, rejection reasons, and Basic versus Advanced demand.
- Improve the employee experience with clearer guidance, progress indicators, and accessible form design.

- Add automated regression tests for duplicate-request boundary cases, approval decisions, and notification recipients.
- Conduct performance tests using a defined test plan and record actual timings and test conditions.

11. APPENDIX

A. Source Code / Configuration

The application is primarily configured on the ServiceNow platform. Include exported update sets, relevant Business Rule scripts, Flow Designer configuration screenshots, or other permitted configuration artifacts if required by the supervisor. Do not include credentials, tokens, or sensitive instance details.

B. Dataset Link

Not applicable to the current workflow as documented: the project uses ServiceNow request and user records rather than an external analytical dataset.

C. GitHub and Project Demo Link

GitHub repository:

https://github.com/nmk2275/ServiceNow-Based-Employee-Laptop-Request-Application.git

ServiceNow demo : demo link